EDG++: Reliability-Aware Electron Density-Enhanced Molecular Geometry Learning

Hongxin Xiang, Xin Jin, Jun Xia, Wenjie Du, Haowen Chen, Stan Z. Li, *IEEE Fellow* and Xiangxiang Zeng, *IEEE Senior Member*

Abstract—Electron density (ED) encodes electronic structure information central to molecular property prediction, yet geometry-based machine learning force fields do not explicitly use it. We propose EDG, a cross-modal distillation framework that transfers ED knowledge—represented as multi-view RGB-D images—from an ED-aware teacher to geometry students as privileged information, requiring no ED computation at inference. However, naive distillation assumes uniform teacher reliability, risking negative transfer from erroneous cross-modal predictions. We extend EDG to EDG++ by introducing a pre-trained reliability estimator and an adaptive global-local thresholding mechanism for selective distillation. We provide theoretical analysis (Proposition 1) establishing a sufficient condition under which selective distillation reduces an upper bound on gradient noise—a condition characteristic of heavy-tailed teacher error distributions in cross-modal settings—and show that the bound is conservative under the positive dependence expected in practice. Experiments on QM9 (12 properties $\times$ 3 architectures) and rMD17 (10 molecules) show that EDG++ improves 34/36 QM9 task-model combinations (average 3.2–9.0%), reduces tail prediction errors by 27–38% on the hardest 1% of test samples, and improves energy/force predictions on 17/20 rMD17 tasks.

Index Terms—Electron density, molecular geometry learning, knowledge distillation, privileged information, reliability-aware learning, machine learning force fields.

1 Introduction

Machine learning force fields (MLFF) can predict molecular energies and forces with near-quantum accuracy at a fraction of the cost of ab initio methods [1]–[5]. Their success stems largely from geometric deep learning architectures that respect E(3) symmetries [6], [7], representing atoms as nodes in geometric graphs [8], [9]. However, existing methods operate at the atomic level—modeling coordinates, multi-body interactions, *etc*. [9]–[11]—without explicitly incorporating information at the electronic level.

Electron density (ED) describes how electrons are distributed within a molecule and is central to predicting quantum chemical properties [12], [13]. *ED is not merely another view of geometry, but the physical manifestation of chemical bonding, polarization, conjugation, and lone-pair effects; many quantum properties are direct functions of the electronic structure.* Nuclear positions dictate the ED, which in turn determines observables such as total energy, forces, and electrostatic potentials. In density functional theory (DFT) [14], the Hohenberg-Kohn theorem [15] formalizes this: the ground-state energy is a unique functional of the electron density.

From an information-theoretic perspective, ED acts as an *approximate latent mediator*: geometry $\rightarrow$ electron density $\rightarrow$ property. The Hohenberg-Kohn theorem supports this view by establishing ground-state energy as a unique functional of the electron density under fixed composition and charge. Standard geometry GNNs implicitly learn this mediating relationship from data, but may lack sufficient inductive bias. Our approach explicitly provides the mediator's structure through DFT-computed ED during training, then distills this knowledge back to geometry models. A further challenge is the symmetry gap between modalities: geometry GNNs enforce E(3) equivariance, whereas ED images from 2D projections of 3D density fields rely on multi-view rendering to approximate rotational invariance. Our distillation framework addresses this: *we transfer electronic structure knowledge from a non-equivariant visual teacher to equivariant geometry students, so that they acquire electron-level inductive bias while retaining symmetry guarantees at inference*.

The application of ED faces three major challenges:

Challenge 1: High computational complexity of ED. Unlike the number of atoms in a molecule, which is usually on the scale of tens to hundreds, the ED relies on a continuous spatial distribution and as the resolution increases, the number of data points may reach millions or even tens of millions. As shown in Fig. 1(a), there are two direct ways to represent ED: point cloud [16] and voxel [17]. As shown in Fig. 1(b), we empirically show the limitations of point clouds and voxels as ED representations in energy

- *H. Xiang is with the College of Computer Science and Electronic Engineering, Hunan University, Changsha, China.*
- *X. Jin is with the Eastern Institute of Technology, Ningbo, China.*
- *J. Xia is with the School of Engineering, Westlake University, Hangzhou, China.*
- *W. Du is with the University of Science and Technology of China, Hefei, China.*
- *H. Chen is with the College of Computer Science and Electronic Engineering, Hunan University, Changsha, China.*
- *S. Li is with the School of Engineering, Westlake University, Hangzhou, China.*
- *X. Zeng is with the College of Computer Science and Electronic Engineering, Hunan University, Changsha, China. (Corresponding author. E-mail: xzeng@hnu.edu.cn)*

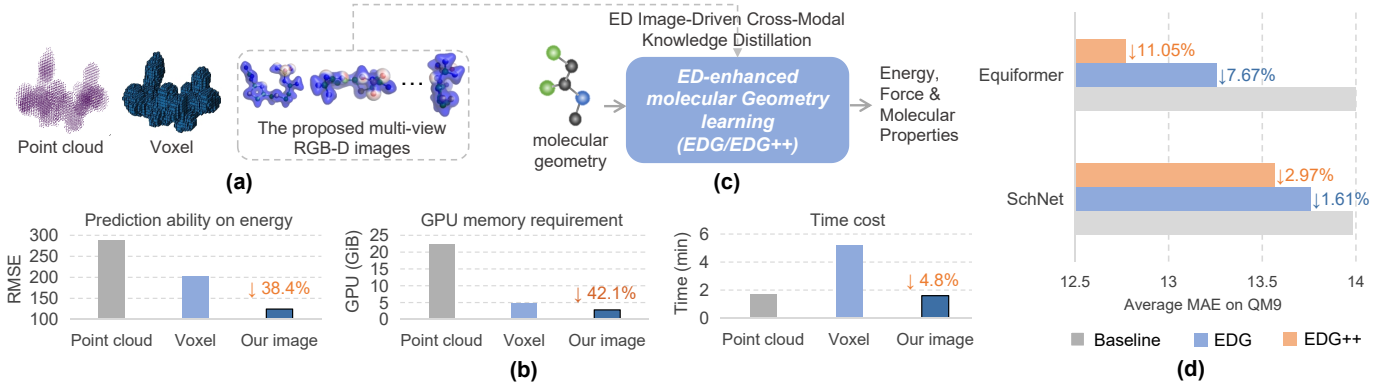

Fig. 1: Motivation and overview (detailed pipeline in Fig. 2). (a) ED representation methods. (b) RMSE, GPU memory, and training time of different ED representations. (c) High-level EDG/EDG++ framework: multi-view ED images with RGB-D channels enhance molecular geometry learning; EDG++ adds reliability-aware selective distillation. (d) Average MAE on 12 QM9 tasks. EDG improves both SchNet and Equiformer; EDG++ further improves through reliability-aware selective distillation.

prediction, GPU memory efficiency, and training efficiency. In particular, point clouds and voxels are directly related to the resolution of the ED, so the computational efficiency will decrease as the resolution increases. These limitations motivate us to propose a multi-view RGB-D image (the right subfigure of Fig. 1(a)) for accurate and efficient ED representation, which produces a fixed-size output regardless of ED grid resolution and compresses the continuous ED signal into pixels. Compared with point clouds and voxels, the proposed images improve energy prediction ability, GPU memory efficiency, and training efficiency by 38.4%, 42.1%, and 4.8%, respectively.

Challenge 2: Expensive ED acquisition. Obtaining ED data requires either costly experimental techniques (*e.g.*, X-ray crystallography [18], neutron diffraction [19]) or computationally intensive quantum chemical calculations (*e.g.*, DFT [14], [20], [21]), both of which are resource-intensive. As shown in Fig. 1(c), we address this by transforming ED-enhanced geometry learning into a teacher-student paradigm. We first use DFT to obtain 2 million high-quality ED data and train an ED representation learning model (ImageED). We then transfer the knowledge in ImageED to an ED-aware teacher that takes structural images (without ED) as input, which subsequently distills geometry students. This scheme, called **EDG** (Electron Density-enhanced Geometry learning), requires ED computation only once during pre-training, eliminating the need for ED data in all subsequent stages. As shown in Fig. 1(d), geometry students equipped with EDG and EDG++ achieve significant prediction improvements on QM9.

Challenge 3: Unreliable teacher predictions causing negative transfer. Beyond feasibility and efficiency, a critical question remains: *when should we trust the teacher's predictions?* The quality of cross-modal predictions varies across samples due to intrinsic DFT approximations, molecular conformation diversity, and information loss when compressing 3D electron clouds into 2D images. *Naive distillation implicitly assumes uniform teacher fidelity, risking negative transfer* [22]. To address this, we propose **EDG++**, a reliability-aware extension that introduces a pre-trained reliability estimator and an adaptive global-local thresholding mechanism for selective distillation, filtering unreliable teacher predictions before knowledge transfer (Section 3.6–3.8).

We summarize the main contributions as follows:

- We propose EDG, an electron density distillation framework that injects electronic structure inductive bias into geometry-based models without adding inference cost, using electron density as privileged information available only during training.
- We design an efficient multi-view RGB-D image representation for electron density and develop ImageED, a masked autoencoder that learns ED-related features, along with an ED-aware teacher that transfers this knowledge without requiring ED at inference.
- We extend EDG to EDG++ by introducing a pre-trained reliability estimator and an adaptive thresholding mechanism combining global and local statistics, enabling selective distillation to mitigate negative transfer. We provide a theoretical analysis establishing a sufficient condition under which selective distillation reduces an upper bound on gradient noise, supported by empirical observation of heavy-tailed error structures.
- We demonstrate improvements on the majority of tasks across multiple architectures on QM9 (SchNet, Equiformer, SphereNet) and rMD17 (SphereNet with EDG++; SchNet, ViSNet with EDG), achieving gains on 34/36 QM9 task-model combinations and 17/20 rMD17 energy/force predictions, while characterizing the boundary conditions of the approach.

Difference with Conference Version. Table 1 summarizes the key differences between the IJCAI 2025 conference paper [23] (EDG) and this journal extension (EDG++).

2 Related Work

2.1 Molecular Geometry Representation Learning

Geometric deep learning drives the progress of machine learning force fields (MLFF) [8], [9]. Equivariant neural networks (ENNs) [24] that respect E(3) symmetries are the

TABLE 1: Conference (IJCAI 2025) vs. journal extension.

Aspect	Conference (EDG)	Journal (EDG++)
Distillation	Naive (all samples)	Selective (reliability-aware)
Reliability est.	–	Pre-trained; ~131K params
Thresholding	–	Adaptive global-local
Theory	–	Prop. 1 (gradient noise)
QM9 models	SchNet, SphereNet	+ Equiformer
rMD17 models	SchNet, ViSNet	+ SphereNet (EDG++)
Ablations	λ only	β, κ, tail error, CV
Analysis	Per-property	+ property-type, architecture

dominant paradigm, with methods modeling inter-atomic interactions [11], [25], angular information [8], [26], [27], and higher-order representations [9], [28], [29]. However, all existing geometry-based methods operate solely on nuclear-level information and do not incorporate electronic structure. Our approach is architecture-agnostic and supplements any geometric model with electronic information.

2.2 Electron Density Representation Learning

Existing electron density (ED) representations fall into two categories. Point cloud methods treat all density values as a point set; for example, PointNet [30] has been applied to classify the symmetry of inorganic compounds [31]. Voxel methods discretize the density field on a regular grid; 3D CNNs [32] have been used to predict molecular exchange energy [17] and discover host-molecule guests [33], while 3D-UNet [34] segments reactive sites and classifies substances [35].

Beyond direct ED representation, several works have explored machine learning approaches to predict or bypass electron density computation. Brockherde *et al*. [36] proposed bypassing the Kohn-Sham equations by directly learning the density-to-energy mapping. Nagai *et al*. [37] used neural networks to learn the exchange-correlation functional from molecular electron densities. Sunshine *et al*. [12] predicted chemical properties from GNN-predicted electron densities, demonstrating that learned ED representations carry meaningful chemical information. More recently, Lee *et al*. [38] proposed self-supervised diffusion processes for electron-aware molecular representation learning, and Gao *et al*. [39] developed equivariant non-local electron density functionals. However, these methods either require ED at inference time, focus on predicting ED itself, or learn ED-aware representations without a cross-modal distillation mechanism. In contrast, we represent ED as multi-view RGB-D images and design a representation learning method to extract features, enabling ED knowledge to be transferred to geometry models without requiring ED at inference.

2.3 Knowledge Distillation and Privileged Information

Knowledge distillation (KD) [40], [41] transfers knowledge from a teacher model to a student model, typically by matching soft predictions or intermediate representations. Feature-based distillation methods [42], [43] extend this paradigm by aligning intermediate layer activations. Wang *et al*. [44] provided a comprehensive review. Recent work has recognized that not all teacher knowledge is equally beneficial. In the selective KD line, SelKD [45] formulates selection via optimal transport, and Shi *et al*. [46] further develop this perspective at scale. Studies on teacher calibration [47], [48] and uncertainty-aware teaching [49], [50] have shown that accounting for the teacher's confidence can significantly improve outcomes.

Cross-modal KD is increasingly studied [51], showing that bridging heterogeneous feature spaces requires explicit alignment. Our framework distills between fundamentally different representations—ED images and atomic graphs—where the representational and symmetry gap poses distinct challenges.

Positioning of EDG++. We distinguish four sources of unreliable supervision: (1) *sample difficulty* [52], [53]; (2) *label noise* [54], [55]; (3) *teacher uncertainty* [45], [46]; (4) *cross-modal alignment quality*—where the cross-modal mapping itself is imperfect for certain samples. EDG++ operates in category (4): task labels are clean, but the quality of the cross-modal prediction varies due to information loss in the 3D-to-2D projection. This motivates a reliability estimator that predicts the teacher's reconstruction error, rather than relying on the teacher's own uncertainty. Our work builds upon the LUPI paradigm [56], [57], with electron density as the privileged modality. Additional discussion of selective learning methods is provided in Appendix E.

3 Our Method

3.1 Preliminaries

Background. Electron density (ED) connects molecular geometry to energy and force predictions in molecular force fields (MFF). As the primary output of quantum mechanical calculations, ED provides the theoretical foundation for constructing accurate MFFs. The Hohenberg-Kohn theorem [15] states that the ground-state energy is a unique functional of the electron density: $E = E[\rho]$. However, computing $\rho(\mathbf{r})$ requires expensive quantum chemical calculations, making it impractical at inference time. Our framework addresses this by treating ED as *privileged information*—available during training but removed at inference to preserve scalability.

Notation and Problem Formulation. The molecular geometry and the corresponding ground-truth labels with k prediction tasks are $\{\mathcal{G}_i = (\mathcal{V}_i, \mathcal{E}_i)\}_{i=1}^n$ and $\{y_i\}_{i=1}^n \in \mathbb{R}^k$ respectively, where $\mathcal{V}_i \in \mathbb{R}^{n_i^v \times (3+d_i^v)}$ (n_i^v, 3, d_i^v are the number of atoms, the coordinates, and the feature dimensions) and $\mathcal{E}_i \in \mathbb{R}^{n_i^e \times d_i^e}$ (n_i^e is the number of edges and d_i^e is the feature dimensions of bonds). The corresponding multi-view ED image and structural image are $\mathcal{U} \in \mathbb{R}^{N_v \times 4 \times H \times W}$ and $\mathcal{S} \in \mathbb{R}^{N_s \times 3 \times H \times W}$ respectively, where N_v and N_s denote the number of views for ED images ($N_v = 6$) and structural images ($N_s = 4$), and H, W are the image height and width. 3 and 4 represent RGB and RGB-D channels, respectively.

The framework comprises four stages: (1) pre-train a masked autoencoder (MAE) [58] with ED encoder f_{EDE} and decoder f_{EDD} to learn representations $\mathcal{F}^{\mathcal{U}} \in \mathbb{R}^{d^U}$ from ED images $\mathcal{U}$; (2) pre-train an ED-aware teacher f_S and ED predictor f_{EDP} to map structural images $\mathcal{S}$ to structural features $\mathcal{F}^{\mathcal{S}} \in \mathbb{R}^{d^S}$ and then to ED-related features $\mathcal{F}^{\mathcal{S}\to\mathcal{U}} \in \mathbb{R}^{d^U}$; (3) pre-train a reliability estimator ϕ_{eval} that predicts the reconstruction error of the teacher for each sample; (4) distill a geometry student f_G using the ED-aware teacher, ED predictor, reliability estimator, and mapper f_M with selective distillation.

3.2 Overview of the Method

We propose **EDG** (**E**lectron **D**ensity-enhanced **G**eometry learning) and its reliability-aware extension **EDG++**. As shown in Fig. 2, the framework has four modules: (a) 2 million molecular conformations are processed with DFT to produce multi-view ED images with RGB-D channels (Section 3.3); (b) ImageED learns ED-related representations from these images via two self-supervised tasks (Section 3.4); (c) an ED-aware teacher is trained to predict ED features from structural images by minimizing the discrepancy with ImageED features (Section 3.5); (d) the teacher distills ED knowledge into a geometry student through an ED predictor and mapper (Section 3.9). EDG++ extends the base framework with a theoretical analysis of when selective distillation reduces gradient noise (Section 3.6), a reliability estimator (Section 3.7), and an adaptive thresholding mechanism (Section 3.8).

3.3 Generation of RGB-D Electron Density Images

As shown in Fig. 2(a), we first obtain 2 million molecular conformations from PCQM4Mv2 [59] and use density functional theory (DFT) with the basis set of 6-31G(d,p) and the exchange-correlation functional of B3LYP to generate DFT data for these molecular conformations [60]. For each molecule, the DFT data includes an electrostatic potential (ESP) file and an ED file stored in the form of a three-dimensional grid.

Using PyMol [61], we jointly render the molecular structure and ED information into multi-view RGB-D images $\mathcal{U} \in \mathbb{R}^{6\times4\times224\times224}$. The RGB channels encode the electrostatic potential distribution of the ED surface using a red-white-blue colormap (representing positive, neutral, and negative regions, respectively), while the depth channel captures the spatial layout. Images are rendered from 6 canonical viewpoints to approximate rotational coverage. The detailed rendering pipeline, including all PyMol commands and view configurations, is provided in Appendix B.

Compared to point cloud and voxel representations of ED, this image-based representation produces a fixed-size output tensor regardless of the source ED grid resolution, and compresses the continuous ED signal into a compact form. Empirically, it improves energy prediction by 38.4%, GPU memory efficiency by 42.1%, and training efficiency by 4.8% compared to the best non-image alternatives.

3.4 ED Representation Learning with ImageED

The ED images exhibit two properties suited to self-supervised learning: (1) RGB channels encode the ED distribution while the depth channel encodes spatial layout, giving each pixel a clear physical meaning; (2) ED is spatially continuous, so masked regions can be predicted from context. We exploit both properties in ImageED, an encoder-decoder framework built on ViT-Large/16 [62] from MAE [58], with two complementary tasks—*visible reconstruction* and *masked reconstruction*—that capture local and contextual ED information from 2 million images.

For a given batch (B molecules) of ED images $u \in \mathbb{R}^{B\times N_v\times c\times H\times W}$ (where $N_v, c, H, W = 6, 4, 224, 224$), we first use a view-agnostic patch embed layer with a patch size of $n_p = 16$ to transform u into a pile of multi-view tokens mt:

$$mt_{i,j} = u\big[\,:,:,:,\ n_p\cdot i : n_p\cdot(i+1),\ n_p\cdot j : n_p\cdot(j+1)\big] \tag{1}$$

We define $n_t = H/n_p = W/n_p$ (= 14 for our setting) and $mt = \{mt_{i,j} | i,j \in \{0,\ldots,n_t-1\}\} \in \mathbb{R}^{B\times N_v\times n_t^2\times(n_p^2\times c)}$. Next, we add positional embeddings and expand along the view to get tokens $t \in \mathbb{R}^{B\times(N_v\times n_t^2)\times(n_p^2\times c)}$. We shuffle the token order and randomly mask 75% to obtain masked tokens t^m and unmasked tokens t^u.

The unmasked tokens are input into the ED encoder f_{EDE}:

$$h^u = f_{EDE}(t^u), \quad h^u \in \mathbb{R}^{B\times(0.25N_v n_t^2)\times d_{\mathcal{U}}} \tag{2}$$

where $d_{\mathcal{U}}$ is the encoder's latent dimension. We append learnable mask tokens $\hat{h}^m$ (initialized to zero) to obtain all tokens:

$$h = \pi(h^u \parallel \hat{h}^m) + pos \in \mathbb{R}^{B\times(N_v n_t^2)\times d_{\mathcal{U}}} \tag{3}$$

where π restores the original token order. The ED Decoder f_{EDD} first projects h from $d_{\mathcal{U}}$ to its own latent dimension and then maps back to the patch dimension $(n_p^2\times c)$, yielding predicted tokens $\hat{t}^m$ and $\hat{t}^u$.

To optimize f_{EDE} and f_{EDD}, we define a *visible reconstruction* task $\mathcal{L}_{VR}$ that reconstructs the unmasked tokens as an auxiliary self-supervision signal:

$$\mathcal{L}_{VR} = \frac{1}{B}\sum_{i=1}^{B} d(t_i^u, \hat{t}_i^u) \tag{4}$$

where $d(\cdot,\cdot)$ denotes the Euclidean distance. Using visible reconstruction alone limits the learning of local features. We therefore add a *masked reconstruction* task $\mathcal{L}_{MR}$ (the standard MAE objective [58]) that reconstructs the masked tokens:

$$\mathcal{L}_{MR} = \frac{1}{B}\sum_{i=1}^{B} d(t_i^m, \hat{t}_i^m) \tag{5}$$

Finally, the overall loss function of ImageED is formulated:

$$\mathcal{L}_{ImageED} = \lambda_{VR}\mathcal{L}_{VR} + \lambda_{MR}\mathcal{L}_{MR} \tag{6}$$

where λ_{VR} and λ_{MR} are the balance coefficients and we set them to 1. After pre-training ImageED with 2 million molecules, we use the f_{EDE} to extract ED features.

3.5 Pretraining of ED-aware Teacher

Since computing ED for every downstream task is prohibitive, we learn the mapping from conformations directly to ED features, bypassing explicit ED computation. The generation path is: conformation $\mathcal{S} \xrightarrow{\text{DFT}}$ ED data $\mathcal{U} \xrightarrow{\text{ImageED}}$ ED features $\mathcal{F}^{\mathcal{U}}$. We adopt a Markov assumption on this chain (*i.e.*, $\mathcal{F}^{\mathcal{U}}$ depends on $\mathcal{S}$ only through $\mathcal{U}$), which justifies learning $p(\mathcal{F}^{\mathcal{U}}|\mathcal{S})$ without ED data at inference (see Appendix C). Under the Born-Oppenheimer framework for ground-state systems with fixed composition and charge, the Hohenberg-Kohn theorem guarantees that the electron density is uniquely determined by the nuclear geometry,

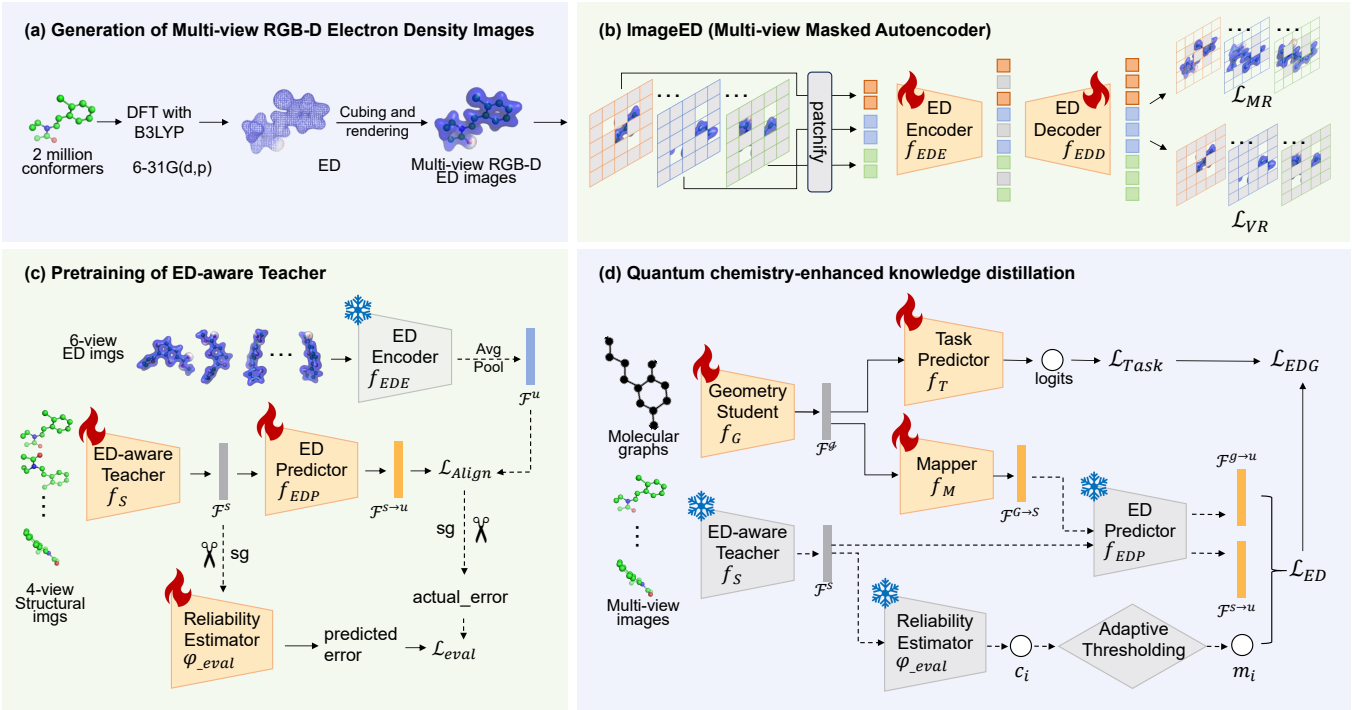

Fig. 2: Overview of the proposed EDG/EDG++ framework. **(a)** Multi-view ED images with RGB-D channels are generated based on 2 million molecular conformers and DFT data, which contains the structural and ED information of the molecule. **(b)** ImageED with masked autoencoder (MAE) architecture and 2 pretext tasks ($\mathcal{L}_{VR}$ and $\mathcal{L}_{MR}$) is pretrained to extract ED-related features from multi-view ED images in (a). **(c)** The ED-aware teacher accepts the structural images as input and learns to transform it into ED features $\mathcal{F}^{\mathcal{S}\to\mathcal{U}}$ using ED predictor and ED encoder in (b). The reliability estimator ϕ_{eval} is trained to predict the reconstruction error. **(d)** In downstream tasks, the ED-aware teacher, ED predictor, and reliability estimator from (c) are frozen to enhance the geometry student. EDG uses naive distillation while EDG++ uses reliability-aware selective distillation with adaptive thresholding. Note that there is no need for explicit involvement of electron density here because the ED teacher has the ability to extract ED-related information.

supporting this assumption. In practice, the lossy multi-view projection from 3D density fields to 2D images introduces residual dependence that the Markov assumption does not capture; the consistent improvements across four architectures on two datasets suggest that this approximation is adequate.

We instantiate this mapping with an ED-aware teacher $f_{\mathcal{S}}$ and ED predictor f_{EDP}. The teacher uses ResNet18 [63] with view-wise average pooling to extract structural features from 4-view structural images $\mathcal{S}$ rendered via PyMol. The ED predictor f_{EDP} is a 2-layer MLP (Linear→Softplus→Linear) that maps structural features to ED features:

$$\mathcal{F}^{\mathcal{S}\to\mathcal{U}} = f_{EDP}(\mathcal{F}^{\mathcal{S}}); \mathcal{F}^{\mathcal{S}} = f_{\mathcal{S}}(s) \tag{7}$$

We freeze the pre-trained ED encoder and use token-wise average pooling to obtain ED features $\mathcal{F}^{\mathcal{U}}$ as the training target. The alignment loss is:

$$\mathcal{L}_{align} = L1(\mathcal{F}^{\mathcal{S}\to\mathcal{U}}, \mathcal{F}^{\mathcal{U}}) \tag{8}$$

where $L1$ denotes L1 distance. With the ED-aware teacher, the costly ED image can be replaced by a cheaper structural image, eliminating DFT-related costs in downstream tasks.

3.6 Theoretical Analysis of Selective Distillation

We analyze why selective distillation can outperform naive distillation when teacher reliability varies across samples. This analysis motivates the reliability estimator (Section 3.7) and the adaptive thresholding mechanism (Section 3.8). The analysis uses simplifying assumptions and yields an upper-bound result; the empirical improvements (Section 4) provide complementary support.

Noisy Teacher Model. In cross-modal distillation, the teacher's ED feature prediction for sample i can be decomposed as:

$$\mathcal{F}_i^{\mathcal{S}\to\mathcal{U}} = \mathcal{F}_i^{\mathcal{U}} + \varepsilon_i \tag{9}$$

where $\mathcal{F}_i^{\mathcal{U}} \in \mathbb{R}^{d^U}$ is the ground-truth ED feature from ImageED and $\varepsilon_i \in \mathbb{R}^{d^U}$ is the teacher's reconstruction error. In cross-modal settings, $\|\varepsilon_i\|$ varies substantially across samples: some molecular geometries are faithfully captured by the multi-view projection, while others (*e.g.*, molecules with delocalized electronic structure or symmetric charge distributions) incur significant information loss.

Distillation Gradient Decomposition. Using the squared L2 loss $\|\cdot\|^2$ for analytical tractability (the qualitative conclusions extend to smooth L1 via local quadratic

approximation), the naive distillation gradient for a batch $\mathcal{B}$ decomposes as:

$$\nabla_\theta \mathcal{L}_{ED}^{naive} = \underbrace{\frac{2}{|\mathcal{B}|}\sum_{i\in\mathcal{B}}\left(\mathcal{F}_i^{\mathcal{G}\to\mathcal{U}} - \mathcal{F}_i^{\mathcal{U}}\right)^\top J_i}_{g_{\text{signal}}} \underbrace{-\frac{2}{|\mathcal{B}|}\sum_{i\in\mathcal{B}}\varepsilon_i^\top J_i}_{g_{\text{noise}}} \quad (10)$$

where $J_i = \nabla_\theta \mathcal{F}_i^{\mathcal{G}\to\mathcal{U}}$ is the Jacobian of the student's predicted ED features. The noise term g_{noise} deflects optimization away from the true ED features. In cross-modal distillation, ε_i is generally not zero-mean—the teacher *systematically* errs on certain molecular conformations—causing consistent directional bias. Our upper-bound analysis (Proposition 1) holds regardless of the mean structure of ε_i, as it relies on Cauchy–Schwarz rather than cancellation of cross terms.

Proposition 1 (Gradient Noise Reduction via Selective Distillation)**.** *Let $\mathcal{B}$ be a training batch and $\mathcal{S} \subseteq \mathcal{B}$ be a selected subset that excludes samples with the largest teacher errors $\|\varepsilon_i\|$. Define $\bar{r}_\mathcal{S} = \frac{1}{|\mathcal{S}|}\sum_{i\in\mathcal{S}}\|\varepsilon_i\|^2$ and $\bar{r}_\mathcal{B} = \frac{1}{|\mathcal{B}|}\sum_{i\in\mathcal{B}}\|\varepsilon_i\|^2$ as the mean squared teacher errors. Under the assumptions that (i) $\|\varepsilon_i\|$ and $\|J_i\|_F$ are independent, and (ii) $\|J_i\|_F \le C$ for all i, upper bounds on the expected squared gradient noise satisfy:*

$$\frac{\mathbb{E}\left[\|g_{noise}^{\mathcal{S}}\|^2\right]}{\mathbb{E}\left[\|g_{noise}^{\mathcal{B}}\|^2\right]} \le \frac{|\mathcal{B}|}{|\mathcal{S}|}\cdot\frac{\bar{r}_\mathcal{S}}{\bar{r}_\mathcal{B}} \quad (11)$$

Consequently, selective distillation reduces the expected gradient noise whenever:

$$\frac{\bar{r}_\mathcal{S}}{\bar{r}_\mathcal{B}} < \frac{|\mathcal{S}|}{|\mathcal{B}|} \quad (12)$$

The proof bounds each term under the independence assumption (Appendix D). Proposition 1 assumes access to the true teacher errors $\|\varepsilon_i\|$ for subset selection; in practice, EDG++ uses a learned reliability estimator (Section 3.7) as a proxy. The proposition thus establishes an *achievable* noise reduction bound—the tighter the estimator's ranking correlates with true $\|\varepsilon_i\|$, the closer EDG++ approaches the oracle bound. The downstream improvements across architectures (Section 4) support that this approximation works well in practice.

When is the condition satisfied? Condition (12) holds whenever the discarded samples carry *disproportionately* high noise relative to their count—*i.e.*, when the teacher error distribution is heavy-tailed. Concretely, if filtering the top α fraction of highest-error samples removes more than α fraction of total squared error, selective distillation reduces gradient noise. We emphasize that Proposition 1 establishes an *upper bound* on gradient noise under simplifying assumptions (bounded Jacobians, Cauchy–Schwarz aggregation); it identifies a sufficient condition for noise reduction rather than a tight characterization.

Section 4.3 provides *indirect but supportive* evidence for the heavy-tailed structure. The evidence is indirect because we observe downstream student prediction errors rather than the teacher reconstruction errors $\|\varepsilon_i\|$ in Proposition 1. The two quantities are causally linked: samples with large $\|\varepsilon_i\|$ receive noisier supervision, producing systematically larger prediction errors. This pattern holds across three architectures and four property types (27–38% tail error reduction), supporting the causal interpretation. The independence assumption between $\|\varepsilon_i\|$ and $\|J_i\|_F$ is also conservative: molecular structural complexity simultaneously increases both teacher error and student sensitivity, inducing positive dependence that tightens the bound (Appendix D).

3.7 Reliability Estimator

Following Proposition 1, we introduce a reliability estimator $\phi_{eval} : \mathbb{R}^{d^S} \to \mathbb{R}$ as a practical proxy for the per-sample teacher error $\|\varepsilon_i\|$. It is a lightweight 2-layer MLP (Linear(512, 256)→Softplus→Linear(256, 1)) jointly trained with the ED-aware teacher in Stage II. Both its input features and regression target are detached from the teacher's computation graph to prevent gradient interference: without input detachment, the evaluator's gradient could push the teacher toward making errors more *predictable* rather than more *accurate*; without target detachment, $\mathcal{L}_{eval}$ would back-propagate through $\mathcal{F}^{\mathcal{S}\to\mathcal{U}}$ and alter the alignment loss landscape. The training objective is:

$$\mathcal{L}_{eval} = \mathbb{E}\left[\left|\phi_{eval}\big(\text{sg}(\mathcal{F}^\mathcal{S})\big) - \text{sg}\big(\|\mathcal{F}^{\mathcal{S}\to\mathcal{U}} - \mathcal{F}^\mathcal{U}\|_2\big)\right|\right] \quad (13)$$

where $\text{sg}(\cdot)$ denotes stop-gradient (detach). At inference, we convert the evaluator output to a confidence score $c_i = -\phi_{eval}(\mathcal{F}^{\mathcal{S},(i)})$, where higher c_i indicates higher estimated reliability. The threshold form $\tau = \mu + \kappa\sigma$ (Section 3.8) is inherently scale-invariant, ensuring that κ controls the retention rate in units of standard deviations regardless of the absolute scale of c_i.

3.8 Adaptive Thresholding Mechanism

We implement selective distillation with an adaptive thresholding mechanism combining global (dataset-level) and local (batch-level) statistics. The form $\tau = \mu + \kappa \cdot \sigma$ is *distribution-agnostic*: since μ and σ come from the same confidence score population, a given κ yields comparable filtering strictness regardless of the score distribution's scale or shape. No distributional assumption is required; under approximate normality, the retention rate simplifies to $\Phi(-\kappa)$ (Appendix D).

Local Threshold. For each mini-batch $\mathcal{B}$, we compute:

$$\tau_{local} = \mu_\mathcal{B} + \kappa_{batch} \cdot \sigma_\mathcal{B} \quad (14)$$

where $\mu_\mathcal{B}$ and $\sigma_\mathcal{B}$ are the mean and standard deviation of confidence scores within the batch, and κ_{batch} controls the strictness of filtering.

Global Threshold. We pre-compute statistics over the entire training set:

$$\tau_{global} = \mu_{all} + \kappa_{all} \cdot \sigma_{all} \quad (15)$$

Hybrid Threshold. The final threshold combines both:

$$\tau = \beta \cdot \tau_{local} + (1-\beta) \cdot \tau_{global} \quad (16)$$

where $\beta \in [0, 1]$ balances local adaptivity and global consistency. Setting $\beta = 0$ yields pure global thresholding, $\beta = 1$ yields pure local thresholding, and $\beta = 0.5$ provides a balanced hybrid strategy.

3.9 ED-enhanced Molecular Geometry Learning

During downstream training, the geometry student f_G and the frozen ED-aware teacher extract features $\mathcal{F}^{\mathcal{G}}$ and $\mathcal{F}^{\mathcal{S}}$ from geometry data and structural images, respectively. Here f_G can be any geometry-based model (*e.g.*, SchNet [25], SphereNet [8], Equiformer [11], ViSNet [9]).

A mapper f_M (a single linear layer) projects geometry features into the structural space; the nonlinear transformation is handled by the frozen ED predictor f_{EDP}:

$$\mathcal{F}^{\mathcal{G}\to\mathcal{U}} = f_{EDP}(f_M(\mathcal{F}^{\mathcal{G}})); \mathcal{F}^{\mathcal{S}\to\mathcal{U}} = f_{EDP}(\mathcal{F}^{\mathcal{S}}) \tag{17}$$

EDG: Naive Distillation. In the base framework EDG, we distill knowledge from all samples:

$$\mathcal{L}_{ED}^{naive} = SL1(\mathcal{F}^{\mathcal{G}\to\mathcal{U}}, \mathcal{F}^{\mathcal{S}\to\mathcal{U}}) \tag{18}$$

where $SL1$ represents smooth L1 distance [64].

EDG++: Selective Distillation. Given the adaptive threshold τ, the selection mask $m_i = \mathbf{1}[c_i \geq \tau]$ determines which samples receive distillation supervision. The selective distillation loss is:

$$\mathcal{L}_{ED}^{selective} = \begin{cases} \frac{\sum_i m_i \cdot SL1(\mathcal{F}_i^{\mathcal{G}\to\mathcal{U}}, \mathcal{F}_i^{\mathcal{S}\to\mathcal{U}})}{\sum_i m_i}, & \text{if } \sum_i m_i > 0 \\ 0, & \text{otherwise} \end{cases} \tag{19}$$

When no samples pass the threshold, only the task loss drives optimization.

Total Training Loss. The task predictor f_T outputs predictions $\hat{y} = f_T(\mathcal{F}^{\mathcal{G}})$, and the final loss is:

$$\mathcal{L}_{EDG} = \underbrace{L1(\hat{y}, y)}_{\mathcal{L}_{Task}} + \lambda \mathcal{L}_{ED} \tag{20}$$

where λ controls the distillation strength, and $\mathcal{L}_{ED}$ is either $\mathcal{L}_{ED}^{naive}$ (EDG) or $\mathcal{L}_{ED}^{selective}$ (EDG++).

Inference. At inference time, only the geometry student and task predictor are used. The ED-aware teacher, reliability estimator, and distillation-related components are not required, ensuring no computational overhead compared to baseline geometry models.

4 Experiments

4.1 Experimental Settings

Datasets and evaluation protocol. To pre-train ImageED, the ED-aware teacher, the ED predictor, and the reliability estimator, we select the first 2 million unlabeled molecular conformations and their DFT-computed ED data from the EDBench database [65], generated by Psi4 software [60] with a grid spacing of 0.4 Bohr. In the evaluation stage, we select 12 widely used tasks related to quantum mechanical properties from QM9 [66] and 10 common tasks related to energy/force from revised MD17 (rMD17) [67]. For force prediction, we first predict the molecular energy and use the gradient of each node position as the force, *i.e.*, *force=-torch.autograd.grad(outputs=energy, inputs=positions)* in PyTorch [68]. The dataset split follows Geom3D [69]: 110K for training, 10K for validation, and 11K for testing in QM9; 950 for training, 50 for validation, and 1000 for testing in rMD17. We use mean absolute error (MAE) as the evaluation metric.

Baselines. To evaluate EDG and EDG++, we select geometry-based models with diverse architectures as geometry students. For QM9, we evaluate SchNet [25], Equiformer [11], and SphereNet [8]. For rMD17, we additionally evaluate ViSNet [9] with EDG, and conduct EDG++ experiments on SphereNet. Following [8], [9], [69], we ensure that each baseline is fully trained with the same hyperparameters as in the original papers.

Implementation details. The implementation details of ImageED pre-training, ED-aware teacher pre-training, and reliability estimator training are provided in Appendix R. In the distillation stage, we conduct grid search over the hyperparameters described in Appendix Q. The mapper f_M and task predictor f_T each consist of a single linear layer. We select the best checkpoint based on validation MAE for QM9 and validation force MAE for rMD17.

Hyperparameter search and evaluation protocol. We follow a strict protocol to avoid data leakage: hyperparameter search is performed exclusively on the validation set, while all reported results are evaluated on the held-out test set. For EDG++ results, we report the best result for each metric independently across all hyperparameter configurations (per-task-best), which provides an *optimistic upper bound* on what the method can achieve when task-specific tuning is feasible. For practical deployment, a single shared configuration should be selected; the default-threshold ($\kappa = 0$) configuration already captures the majority of the improvement (Table 6). EDG baselines are evaluated with the best λ from the same grid, ensuring a fair comparison on the hyperparameter search budget.

Single-seed protocol. All experiments use a single random seed (seed=42). We assess statistical reliability via: (1) paired bootstrap CIs on per-sample test predictions (Appendix P); (2) sign tests (34/36 QM9 wins: $p < 10^{-8}$; 17/20 rMD17: $p = 0.001$); and (3) coefficient of variation across 45 configurations (Section 4.5.2). Multi-seed evaluation remains a limitation (Section 5).

4.2 Main Results on QM9

Table 2 presents the main QM9 results. EDG++ improves 34 out of 36 task-model combinations (sign test $p < 10^{-8}$), with average relative gains of 3.2% (SchNet), 9.0% (Equiformer), and 3.6% (SphereNet). These are per-task-best results; the default-threshold ($\kappa = 0$) configuration captures most of the gains (Table 6). Comparing EDG++ with EDG, selective distillation outperforms naive distillation on the majority of tasks (SchNet: 10/12, SphereNet: 7/12), confirming that reliability-aware filtering successfully excludes unreliable teacher predictions.

Property-dependent improvements. The improvements exhibit a physically interpretable hierarchy consistent with the Hohenberg-Kohn theorem: thermodynamic properties achieve the largest and most consistent improvements (5/5 win rate across all architectures; avg. +3.70% to +16.30%), as these ground-state observables are uniquely determined by the electron density [15]. The only two degradation cases—Equiformer on μ (↓1.9%) and ZPVE (↓3.2%)—are architecture-specific and improve on SchNet and SphereNet, confirming this is not a fundamental limitation. Figure 3 visualizes this hierarchy across all 36 combinations; the detailed property-group breakdown is in Appendix G.

TABLE 2: MAE performance on 12 quantum mechanical property prediction tasks in QM9. "Baseline" denotes the original geometry model without distillation. "EDG" denotes naive distillation (conference version). "EDG++" denotes reliability-aware selective distillation; for each property, we report the best result across the hyperparameter search. Δ represents the relative improvement of EDG++ over the baseline. Bold indicates the best result in each column. ↓ indicates lower is better.

	Method	$\alpha \downarrow$ a_0^3	$\Delta\varepsilon \downarrow$ meV	$\varepsilon_{\text{HOMO}} \downarrow$ meV	$\varepsilon_{\text{LUMO}} \downarrow$ meV	$\mu \downarrow$ D	$C_v \downarrow$ $\frac{cal}{mol \cdot K}$	$G \downarrow$ meV	$H \downarrow$ meV	$R^2 \downarrow$ a_0^2	$U \downarrow$ meV	$U_0 \downarrow$ meV	ZPVE ↓ meV
SchNet	Baseline	0.07021	50.829	31.952	26.168	0.03013	0.03228	14.678	14.090	0.13455	14.142	13.915	1.714
	EDG	0.06866	49.778	31.884	25.972	**0.02980**	0.03162	14.022	13.841	**0.12458**	13.794	13.826	1.688
	EDG++	**0.06810**	**49.426**	**31.385**	**25.499**	0.02984	**0.03147**	**13.729**	**13.773**	0.12591	**13.613**	**13.416**	**1.674**
	Δ	↑3.0%	↑2.8%	↑1.8%	↑2.6%	↑1.0%	↑2.5%	↑6.5%	↑2.2%	↑6.4%	↑3.7%	↑3.6%	↑2.3%
Equiformer	Baseline	0.06762	46.308	26.017	23.681	0.02074	0.02733	18.439	16.453	0.45828	15.339	23.928	1.537
	EDG	0.06476	45.813	25.492	23.266	**0.01985**	0.02642	15.976	**14.451**	0.43947	15.466	**16.517**	**1.529**
	EDG++	**0.06175**	**45.089**	**25.269**	**22.283**	0.02113	**0.02637**	**12.058**	14.709	**0.40354**	**14.802**	16.918	1.586
	Δ	↑8.7%	↑2.6%	↑2.9%	↑5.9%	↓1.9%	↑3.5%	↑34.6%	↑10.6%	↑11.9%	↑3.5%	↑29.3%	↓3.2%
SphereNet	Baseline	0.04670	40.129	22.007	19.435	0.02689	0.02437	7.875	7.199	0.25821	6.999	6.641	1.253
	EDG	0.04592	39.694	21.842	19.014	**0.02648**	**0.02376**	7.769	**6.283**	**0.24935**	6.502	**6.101**	1.206
	EDG++	**0.04567**	**39.470**	**21.766**	**18.750**	0.02652	0.02400	**7.656**	6.364	0.25341	**6.409**	6.424	**1.198**
	Δ	↑2.2%	↑1.6%	↑1.1%	↑3.5%	↑1.4%	↑1.5%	↑2.8%	↑11.6%	↑1.9%	↑8.4%	↑3.3%	↑4.4%

Avg. Δ: SchNet ↑3.2%, Equiformer ↑9.0%, SphereNet ↑3.6%. Overall 34/36 tasks improved.

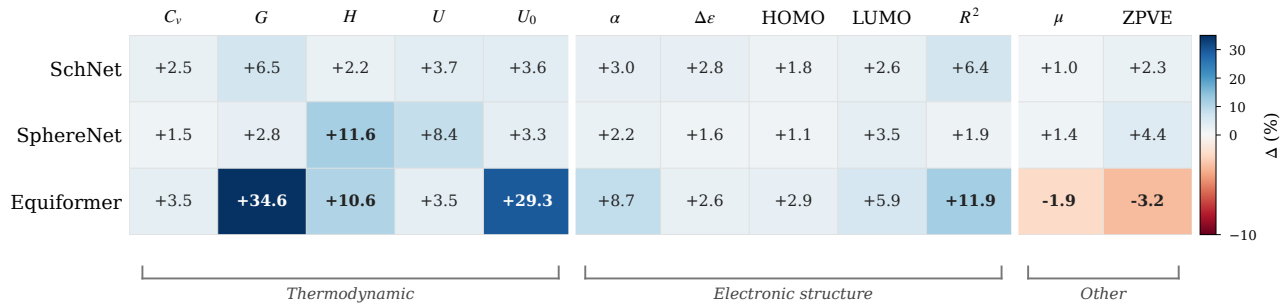

Fig. 3: Relative improvement (%) for all 36 QM9 task-model combinations (3 architectures × 12 properties). Properties are grouped by Hohenberg-Kohn relevance: thermodynamic (left), electronic (middle), and other (right). Blue cells indicate improvement; warm-colored cells indicate degradation.

4.3 Tail Error and Robustness Analysis

To test the heavy-tailed error structure assumed in Proposition 1, we analyze how threshold tuning affects prediction errors across difficulty regimes. We compare two EDG++ configurations—tuned κ versus default κ=0—on four representative tasks spanning different architectures and property types.

Figure 4 shows that threshold tuning substantially reduces tail errors: at the top 1%, reductions range from 27.0% (SphereNet-H) to 37.9% (Equiformer-U_0), providing indirect evidence for the heavy-tailed error structure assumed in Proposition 1 (see Section 3.6 for the causal link between downstream prediction errors and teacher reconstruction errors). The tuned curve remains below the default curve across the full visible range (top 0.5%–25%), indicating that the benefit extends broadly across the harder half of the test set. Standard deviation reductions of 5.3–17.3% and maximum error reductions up to 39.1% (Equiformer-U_0) confirm improved robustness. Difficulty-stratified analysis (Appendix F) reveals a capacity reallocation effect: threshold tuning shifts learning capacity from easy samples (where the baseline already performs well) to hard samples (where stricter filtering provides the most corrective signal), consistent with Proposition 1.

4.4 Main Results on rMD17

Tables 3 and 4 present the rMD17 results. EDG achieves consistent energy improvements across all architectures (SchNet: 33.7%, SphereNet: 25.2% average), with more modest force improvements (1.5–5.3%), as force depends on local atomic interactions rather than the global energy distribution captured by ED. Note that EDG++ experiments on rMD17 are conducted on SphereNet only due to computational constraints; the cross-architecture generality of EDG++ is established on QM9 (three architectures), while rMD17 provides a single-architecture case study with molecule-level analysis. We note that rMD17 uses only 50 validation samples, so per-molecule threshold tuning carries higher overfitting risk than on QM9.

Progressive improvement from EDG to EDG++. To isolate the reliability evaluator's contribution, we introduce a controlled baseline: EDG ($\lambda = 0.001$) applies naive distillation with the same λ as EDG++ but *without* the reliability estimator. Comparing EDG ($\lambda = 0.001$) → EDG++ ($\kappa = 0$), which differs *only* in whether the evaluator is active, the evaluator improves energy on 8/10 molecules, with naphthalene (↓54.5%), salicylic acid (↓32.4%), and aspirin (↓30.7%) showing the largest gains. Notably, naive distillation at $\lambda = 0.001$ without the evaluator *degrades*

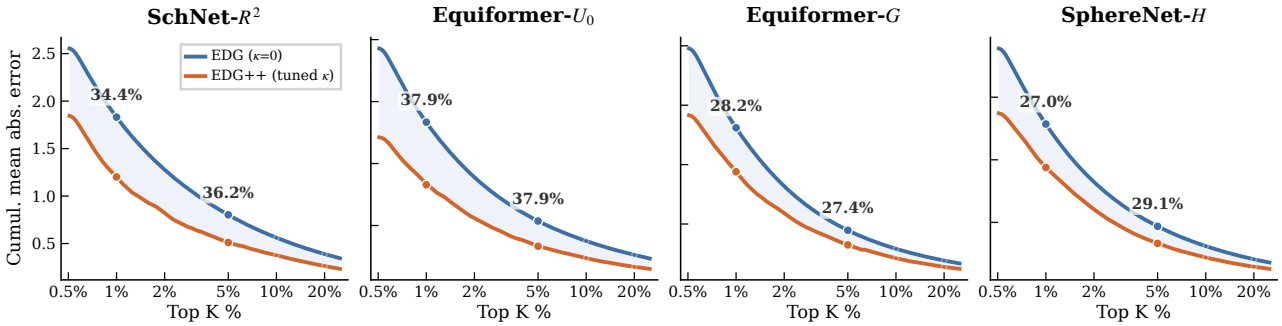

Fig. 4: Cumulative mean absolute error curves on QM9 for four representative tasks. The x-axis (log scale) shows the top K% of samples ranked by difficulty (default-threshold error magnitude), from the hardest 0.5% to 25%. The EDG++ tuned-κ curve (orange) lies consistently below the EDG++ default-threshold ($\kappa = 0$) curve (blue). Annotated percentages show the relative error reduction at the 1% and 5% marks. *Note*: both curves use the reliability estimator; the comparison isolates the effect of threshold tuning. Difficulty ranking is defined by the default-threshold model's errors; y-axis scales differ across panels to maximize readability within each task.

TABLE 3: MAE performance on 10 energy ($\frac{kcal}{mol}$) prediction tasks in rMD17. Baseline and EDG results for all architectures are from the conference version [23]. EDG++ experiments are conducted on SphereNet. "EDG ($\lambda = 0.001$)" denotes naive distillation with fixed $\lambda = 0.001$ and no reliability estimator, serving as a controlled baseline to isolate the evaluator's contribution. "EDG++ ($\kappa = 0$)" denotes the default-threshold configuration (reliability estimator active, $\kappa_{all} = 0$, $\lambda = 0.001$). "EDG++" denotes the best κ_{all} for each molecule. Bold indicates the best result within each model block. Malona. and Naphth. abbreviate Malonaldehyde and Naphthalene.

	Method	Aspirin ↓	Azobenz. ↓	Benzene ↓	Ethanol ↓	Malona. ↓	Naphth. ↓	Paracet. ↓	Salicylic ↓	Toluene ↓	Uracil ↓
SchNet	Baseline	0.73909	0.39678	0.02052	0.12516	0.16142	0.21158	0.37097	0.19078	0.20797	0.07872
	EDG	0.35525	0.33441	0.01711	0.06061	0.11181	0.07338	0.28303	0.15697	0.08780	0.07433
SphereNet	Baseline	0.18091	0.09794	0.00647	0.03784	0.06005	0.03823	0.10425	0.14119	0.03452	0.08088
	EDG	0.13622	0.06788	0.00413	**0.03575**	**0.05659**	0.02753	0.09934	0.09569	0.02413	**0.03683**
	EDG ($\lambda = 0.001$)	0.18166	**0.06603**	0.00377	0.04695	0.05806	0.05216	0.10866	0.09713	0.03175	0.05925
	EDG++ ($\kappa = 0$)	0.12584	0.07125	**0.00341**	0.04081	0.06767	0.02373	0.09734	**0.06563**	0.02367	0.05570
	EDG++	**0.12391**	0.07125	**0.00341**	**0.03575**	0.06611	**0.02256**	**0.09628**	**0.06563**	**0.02220**	0.04841
ViSNet	Baseline	0.05547	0.02081	0.00627	0.01095	0.01517	0.01313	0.02700	0.01966	0.01089	0.01238
	EDG	**0.04650**	**0.01838**	**0.00616**	**0.00990**	**0.01395**	**0.01178**	**0.02491**	**0.01906**	**0.00998**	**0.01188**

TABLE 4: MAE performance on 10 force ($\frac{kcal}{mol \cdot \text{Å}}$) prediction tasks in rMD17. Setup follows Table 3.

	Method	Aspirin ↓	Azobenz. ↓	Benzene ↓	Ethanol ↓	Malona. ↓	Naphth. ↓	Paracet. ↓	Salicylic ↓	Toluene ↓	Uracil ↓
SchNet	Baseline	1.04245	0.90082	0.18569	0.38519	0.65536	0.39851	0.82544	0.77487	0.48322	0.51399
	EDG	1.04910	0.91692	0.17113	0.37901	0.64678	0.39509	0.83296	0.74308	0.47790	0.50783
SphereNet	Baseline	0.39134	0.21776	0.02151	0.19432	0.29278	0.11141	0.32265	0.28692	0.10978	0.27702
	EDG	0.38598	**0.21665**	0.02101	0.18741	**0.28456**	0.11102	0.32023	**0.28299**	0.10804	**0.17179**
	EDG ($\lambda = 0.001$)	0.38870	0.21670	0.02160	**0.18594**	0.29120	0.11220	**0.31742**	0.28919	0.10706	0.26561
	EDG++ ($\kappa = 0$)	0.37960	0.22238	0.02078	0.18871	0.31599	0.09711	0.32309	0.29871	**0.10341**	0.23117
	EDG++	**0.37929**	0.21708	**0.02012**	0.18698	0.31599	**0.09541**	0.31846	0.29452	**0.10341**	0.22544
ViSNet	Baseline	0.15164	0.05729	0.00656	0.05688	0.09275	0.02808	0.10488	0.08348	0.02980	0.05252
	EDG	**0.14996**	**0.05691**	**0.00647**	**0.05558**	**0.08992**	**0.02798**	**0.10599**	**0.08107**	**0.02780**	**0.05100**

energy below baseline on 4/10 molecules (e.g., naphthalene: ↑36.4%); the evaluator recovers 3 of these 4 cases to below-baseline error, demonstrating that reliability-aware filtering is essential for safe distillation at this weight. Threshold tuning (EDG++ $\kappa = 0 \rightarrow$ EDG++ best κ) provides further incremental gains. Overall, EDG++ achieves the best energy on 7/10 and best force on 4/10 molecules; the smaller force gains are expected, as force depends on local atomic interactions rather than the global energy distribution captured by ED.

Malonaldehyde: a boundary case. Malonaldehyde is the only molecule where the reliability estimator consistently degrades performance. Instructively, naive distillation at $\lambda = 0.001$ *does* improve energy (0.05806 vs. baseline 0.06005, ↓3.3%), indicating that the ED-based distillation signal is beneficial. However, activating the evaluator (EDG++ $\kappa = 0$: 0.06767) reverses this gain, suggesting that the evaluator incorrectly filters beneficial distillation signal for this molecule—likely because malonaldehyde's rapid intramolecular proton transfer dynamics produce ED patterns that the evaluator misclassifies as unreliable. EDG with per-task-best λ achieves the best result (0.05659), and threshold tuning partially mitigates the degradation (0.06611 vs. 0.06767 at κ=0), pointing to molecule-adaptive thresholding as a future direction. A per-sample error visualization is

TABLE 5: Ablation on thresholding strategy (β) for QM9. We report the number of tasks (out of 12) where each β value achieves the best result. "Total" aggregates across all three architectures (out of 36).

Strategy	β	SchNet	SphereNet	Equiformer	Total
Global only	0	2	3	7	12/36
Global+Local	0.5	5	7	5	**17/36**
Local only	1.0	5	2	0	7/36

TABLE 6: Incremental contribution of threshold tuning on QM9. "$\kappa = 0$" denotes the default-threshold configuration. Δ_{inc} is the incremental improvement from tuning κ beyond the default.

Model	Improved	Avg. Δ_{inc}	Max Δ_{inc}
SchNet	10/12	+0.97%	+2.46% (R^2)
SphereNet	9/12	+1.00%	+3.00% (ZPVE)
Equiformer	8/12	+2.87%	+12.82% (H)

provided in Appendix I; additional sensitivity analysis is in Appendix J.

4.5 Ablation Studies

4.5.1 Selective Distillation Components

Table 5 analyzes the mixing coefficient β. The hybrid strategy ($\beta = 0.5$) achieves the most wins overall (17/36). The preference is architecture-dependent: Equiformer favors global thresholding (7/12), while SchNet and SphereNet benefit from hybrid or local strategies, suggesting that architectures with global attention already capture batch-level variation internally.

Table 6 isolates the contribution of threshold tuning by comparing tuned κ against the default κ=0 configuration. Even with $\kappa = 0$, the reliability estimator is active and filters samples below the mean reliability score; tuning κ provides consistent incremental improvements across architectures (avg. +0.97% to +2.87%). Equiformer benefits the most, consistent with its higher sensitivity to filtering thresholds. The fact that 27–31 out of 36 tasks improve provides indirect but strong evidence for the reliability estimator: if the confidence ranking were uncorrelated with actual teacher error, threshold-based filtering would not produce systematic gains.

4.5.2 Hyperparameter Sensitivity

The distillation weight λ exhibits architecture-dependent sensitivity (detailed per-task results in Appendix H): Equiformer averages 7.77% sensitivity (5× SchNet's 1.47%), with thermodynamic properties most sensitive (U_0: 34.87%). SchNet maintains consistently low sensitivity (max 2.37%). The coefficient of variation (CV) of MAE across all 45 configurations further confirms robustness: SchNet (mean CV = 1.1%), SphereNet (1.8%), and Equiformer (4.21%). Even Equiformer's default-threshold (κ=0) captures the majority of the improvement (Table 6). Additional hyperparameter interaction analysis is in Appendix J.

4.6 Advantages of ED Image Representation

As shown in Fig. 1(b), the proposed multi-view RGB-D image representation outperforms point cloud and voxel alternatives on all 6 energy prediction tasks (7.8–71.6% relative RMSE gains on 10,000 molecules with an 8:1:1 split), while producing a fixed-size output independent of source grid resolution. Detailed per-task RMSE values are provided in Table S3 (Appendix L). Qualitative visualization of ImageED reconstruction quality is provided in Appendix K.

5 Conclusion

We have presented EDG and EDG++, a framework that uses electron density as privileged information to improve geometry-based molecular models through cross-modal distillation. By representing ED as multi-view RGB-D images and training an ED-aware teacher, the approach eliminates quantum chemical calculations at inference. EDG++ adds reliability-aware selective distillation—motivated by a gradient noise analysis (Proposition 1) identifying sufficient conditions for noise reduction—to avoid negative transfer from unreliable teacher predictions. Experiments show improvements on 34/36 QM9 task-model combinations and 17/20 rMD17 energy/force predictions, with the improvement hierarchy aligning with the Hohenberg-Kohn theorem: thermodynamic properties benefit most, followed by electronic structure properties.

Limitations. (1) All results are reported with a single random seed (seed=42). While the sign test ($p < 10^{-8}$ for 34/36 QM9 wins) and bootstrap CI analysis (Appendix P) provide evidence that the improvements are consistent, multi-seed evaluation would strengthen the conclusions. (2) EDG++ results are reported per-task-best, which provides an optimistic upper bound; the default-threshold configuration captures the majority of gains (Table 6). (3) EDG++ on rMD17 is evaluated on SphereNet only; cross-architecture generality on rMD17 remains to be established. (4) A non-ED teacher control is not included; while indirect evidence supports that gains originate from the ED signal (see Appendix N), a formal ablation would be valuable. (5) The reliability estimator is frozen during downstream tasks; joint fine-tuning may yield further improvements.

Future Directions. Several directions merit investigation: soft weighting schemes that modulate distillation strength continuously, molecule-adaptive distillation weights for conformationally challenging systems, and extension to other privileged information sources such as molecular orbitals or experimental spectra. The scale-invariant threshold $\tau = \mu + \kappa\sigma$ and the reliability-aware filtering mechanism can serve as general components for cross-modal knowledge transfer beyond molecular learning.

Code Availability. The source code and pretrained models are available at https://github.com/HongxinXiang/EDG.

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APPENDIX A
THE OVERALL PROCESS OF EDG/EDG++

In order to clearly describe the proposed EDG/EDG++ framework, we show the overall process in Algorithm 1.

Algorithm 1 The overall process of EDG/EDG++ framework.

Input: Molecular geometries $\mathcal{G}$, ground-truth labels y; multi-view ED images $\mathcal{U}$, structural images $\mathcal{S}$.

Stage I: Pre-training ImageED
Train ED encoder f_{EDE} and decoder f_{EDD} on 2M ED images using:
$\mathcal{L}_{ImageED} = \lambda_{VR}\mathcal{L}_{VR} + \lambda_{MR}\mathcal{L}_{MR}$
Freeze f_{EDE} after pre-training.

Stage II: Pre-training ED-aware Teacher and Reliability Estimator
Jointly train ED-aware teacher $f_{\mathcal{S}}$, ED predictor f_{EDP}, and reliability estimator ϕ_{eval} (EDG++ only):
$\mathcal{F}^{\mathcal{S}} = f_{\mathcal{S}}(\mathcal{S})$, $\mathcal{F}^{\mathcal{S}\to\mathcal{U}} = f_{EDP}(\mathcal{F}^{\mathcal{S}})$
$\mathcal{L}_{align} = L1(\mathcal{F}^{\mathcal{S}\to\mathcal{U}}, \mathcal{F}^{\mathcal{U}})$
$\mathcal{L}_{eval} = \mathbb{E}\big[\big|\phi_{eval}(\text{detach}(\mathcal{F}^{\mathcal{S}})) - \text{detach}(\|\mathcal{F}^{\mathcal{S}\to\mathcal{U}} - \mathcal{F}^{\mathcal{U}}\|_2)\big|\big]$
// detach(): stop gradient; ϕ_{eval} receives no gradient from $f_{\mathcal{S}}$ or f_{EDP}
$\mathcal{L}_{StageII} = \mathcal{L}_{align} + \mathcal{L}_{eval}$
Freeze $f_{\mathcal{S}}$, f_{EDP}, and ϕ_{eval} after pre-training.

Stage III: Downstream Task Training
for each mini-batch $(\mathcal{G}, \mathcal{S}, y)$ **do**
Extract geometry features: $\mathcal{F}^{\mathcal{G}} = f_G(\mathcal{G})$
Extract structural features: $\mathcal{F}^{\mathcal{S}} = f_{\mathcal{S}}(\mathcal{S})$ (frozen)
Compute ED features: $\mathcal{F}^{\mathcal{G}\to\mathcal{U}} = f_{EDP}(f_M(\mathcal{F}^{\mathcal{G}}))$
$\mathcal{F}^{\mathcal{S}\to\mathcal{U}} = f_{EDP}(\mathcal{F}^{\mathcal{S}})$

// EDG++: Selective Distillation
Compute confidence: $c_i = -\phi_{eval}(\mathcal{F}^{\mathcal{S},(i)})$
Local threshold: $\tau_{local} = \mu_{\mathcal{B}} + \kappa_{batch} \cdot \sigma_{\mathcal{B}}$ ($\mu_{\mathcal{B}}, \sigma_{\mathcal{B}}$: batch mean/std of $\{c_i\}$)
Global threshold: $\tau_{global} = \mu_{all} + \kappa_{all} \cdot \sigma_{all}$ (pre-computed over training set)
Hybrid threshold: $\tau = \beta \cdot \tau_{local} + (1-\beta) \cdot \tau_{global}$
Selection mask: $m_i = \mathbf{1}[c_i \geq \tau]$

Compute losses:
$\mathcal{L}_{Task} = L1(\hat{y}, y)$
if $\sum_i m_i > 0$ **then**
$\mathcal{L}_{ED} = \frac{\sum_i m_i \cdot SL1(\mathcal{F}_i^{\mathcal{G}\to\mathcal{U}}, \mathcal{F}_i^{\mathcal{S}\to\mathcal{U}})}{\sum_i m_i}$
else
$\mathcal{L}_{ED} = 0$ *// skip distillation when no sample is selected*
end if
Update f_G, f_M, f_T using $\mathcal{L}_{EDG} = \mathcal{L}_{Task} + \lambda\mathcal{L}_{ED}$
end for

Inference: Only use f_G and f_T; no ED computation required.
return Trained geometry student f_G and task predictor f_T

APPENDIX B
PYMOL RENDERING PIPELINE

This section provides the detailed PyMol [61] commands used for generating multi-view RGB-D electron density images.

Structural Loader. The structural loader uses the command *load {conformation file}* in PyMol to load the structural information from the molecular conformation file.

ED Loader. The ED loader uses the commands *load {ED file}, ED; load {ESP file}, ESP; ramp_new legend, ESP, [-0.08, 0, 0.08], [red, white, blue]; isosurface surface, ED, 0.05; set surface_color, legend, surface* in PyMol to load the ED information, which uses a red-white-blue distribution range to describe the electrostatic potential distribution of ED and red, white, and blue represent positive, neutral, and negative regions, respectively.

Multi-view Joint Render. Multi-view joint render uses commands `set transparency, 0.4; turn {axis}, {angle}; png {path}, width={width}, height={height}` in PyMol to render the ED information and structural information into multi-view RGB-D ED images $\mathcal{U} \in \mathbb{R}^{6\times4\times224\times224}$. Specifically, $\{axis\}$ and $\{angle\}$ represent the rotation angle degrees along axis and we set $(\{axis\}, \{angle\})$ to $(x, 0)$, $(x, 180)$, $(x, 90)$, $(x, -90)$, $(y, 90)$, $(y, -90)$, which means generating images from 6 different views. $\{width\} = 224$ and $\{height\} = 224$ represent the width and height of the rendered image.

Structural Render. The structural render uses the command template *turn {axis}, {angle}; png {path}* in PyMol to render molecular structures from 2 million conformations into multi-view structural images $\mathcal{S}$. We generate 4 views with $(\{axis\}, \{angle\})$ set to $(x, 0), (x, 180), (y, 180), (z, 180)$.

APPENDIX C
MARKOV ASSUMPTION AND PROBABILITY DERIVATION

We provide the full probability derivation for the ED-aware teacher. Let s, x, and h denote the molecular conformation, DFT-computed ED data, and ED features, respectively. The acquisition of ED features follows the path $s \xrightarrow{\text{DFT}} x \xrightarrow{\text{ImageED}} h$.

According to the probability chain rule, the joint probability distribution $p(h, x|s)$ can be decomposed into:

$$p(h, x|s) = p(h|x, s) \cdot p(x|s) \tag{21}$$

According to the Markov hypothesis [70], since h is generated from x via ImageED without additional dependence on s, we have $p(h|x, s) = p(h|x)$. Marginalizing over x:

$$p(h|s) = \int p(h|x) \cdot p(x|s) dx \tag{22}$$

Since both stages of the pipeline are deterministic given their inputs—DFT maps s to a unique x, and ImageED maps x to a unique h—the conditional distributions collapse to point masses: $p(x|s) \approx \delta(x - \text{DFT}(s))$ and $p(h|x) \approx \delta(h - f_{EDE}(x))$. The integral therefore reduces to $p(h|s) \approx \delta(h - f_{EDE}(\text{DFT}(s)))$, which motivates directly learning the composite mapping $s \mapsto h$. The Markov assumption $p(h|x, s) = p(h|x)$ holds exactly when the ImageED encoder is a deterministic function of x alone. In the context of molecular ED, this is well-motivated: under the Born-Oppenheimer approximation for ground-state systems with fixed composition, charge, and spin multiplicity, the Hohenberg-Kohn theorem [15] guarantees that the nuclear geometry uniquely determines the electron density. However, two practical factors introduce residual dependence on s: (1) the DFT computation uses a finite basis set (6-31G(d,p)) and approximate exchange-correlation functional (B3LYP), and (2) the 6-view 2D projection introduces lossy compression of the 3D density field. We therefore treat this as an *approximate* Markov assumption that serves as a practical modeling choice. Its adequacy is validated by the scale of pre-training (2 million molecules) and by the consistent downstream improvements across four architectures and two datasets.

APPENDIX D
PROOF AND REMARKS FOR PROPOSITION 1

D.1 Proof of Proposition 1

Expanding $\|g^{\mathcal{B}}_{\text{noise}}\|^2 = \frac{4}{|\mathcal{B}|^2} \left\|\sum_{i\in\mathcal{B}} \varepsilon_i^\top J_i\right\|^2$:

$$\|g^{\mathcal{B}}_{\text{noise}}\|^2 = \frac{4}{|\mathcal{B}|^2} \left(\sum_{i\in\mathcal{B}} \|\varepsilon_i^\top J_i\|^2 + \sum_{i\neq j} (\varepsilon_i^\top J_i)^\top (\varepsilon_j^\top J_j) \right) \tag{23}$$

Under the independence assumption between $\|\varepsilon_i\|$ and $\|J_i\|_F$, and treating the teacher errors ε_i as fixed for each sample while taking expectation over the student parameters θ (which determine J_i), the cross terms vanish in expectation: for $i \neq j$, the independence of J_i and J_j conditional on $\varepsilon_i, \varepsilon_j$ yields $\mathbb{E}[(\varepsilon_i^\top J_i)^\top (\varepsilon_j^\top J_j)] = 0$. This gives:

$$\mathbb{E}[\|g^{\mathcal{B}}_{\text{noise}}\|^2] = \frac{4}{|\mathcal{B}|^2} \sum_{i\in\mathcal{B}} \mathbb{E}[\|\varepsilon_i^\top J_i\|^2] \leq \frac{4}{|\mathcal{B}|^2} \sum_{i\in\mathcal{B}} \|\varepsilon_i\|^2 \cdot C^2 = \frac{4C^2}{|\mathcal{B}|} \cdot \bar{r}_{\mathcal{B}} \tag{24}$$

where we used $\|\varepsilon_i^\top J_i\|^2 \leq \|\varepsilon_i\|^2 \|J_i\|_F^2$ and the bound $\|J_i\|_F \leq C$, and the last equality follows from $\bar{r}_{\mathcal{B}} = \frac{1}{|\mathcal{B}|} \sum_{i\in\mathcal{B}} \|\varepsilon_i\|^2$. Analogously for the selected subset:

$$\mathbb{E}[\|g^{\mathcal{S}}_{\text{noise}}\|^2] \leq \frac{4C^2}{|\mathcal{S}|} \cdot \bar{r}_{\mathcal{S}} \tag{25}$$

Taking the ratio of the upper bounds:

$$\frac{\mathbb{E}[\|g^{\mathcal{S}}_{\text{noise}}\|^2]}{\mathbb{E}[\|g^{\mathcal{B}}_{\text{noise}}\|^2]} \leq \frac{|\mathcal{B}|}{|\mathcal{S}|} \cdot \frac{\bar{r}_{\mathcal{S}}}{\bar{r}_{\mathcal{B}}} \tag{26}$$

The improvement condition Eq. (13) follows directly: selective distillation reduces gradient noise whenever the fraction of noise variance retained is strictly less than the fraction of samples retained. □

D.2 Conservativeness of the Independence Assumption

The independence between $\|\varepsilon_i\|$ and $\|J_i\|_F$ is adopted for analytical tractability. We argue that relaxing this assumption *strengthens* the conclusion, making the bound conservative.

Physical basis for positive dependence. In cross-modal distillation, molecular structural complexity acts as a confounding variable that simultaneously governs both teacher fidelity and student sensitivity. Structurally challenging molecules—those with conformational flexibility, delocalized electronic structures, or subtle non-covalent interactions—incur larger teacher reconstruction errors $\|\varepsilon_i\|$ (because the 3D-to-2D multi-view projection loses more information for such geometries) and simultaneously produce larger student Jacobian norms $\|J_i\|_F$ (because these molecules occupy high-uncertainty regions of the geometry model's prediction landscape). This shared dependence induces positive correlation, which we formalize as: the conditional expectation $\mathcal{J}(\epsilon) \triangleq \mathbb{E}[\|J_i\|_F^2 \mid \|\varepsilon_i\| = \epsilon]$ is non-decreasing in ϵ.

Formal consequence. Let $\bar{\mathcal{J}}_{\mathcal{A}} = \frac{1}{|\mathcal{A}|}\sum_{i\in\mathcal{A}} \mathcal{J}(\|\varepsilon_i\|)$ denote the mean conditional Jacobian norm over any index set $\mathcal{A}$. Since $\mathcal{S} \subseteq \mathcal{B}$ retains only low-$\|\varepsilon_i\|$ samples and $\mathcal{J}(\cdot)$ is non-decreasing, we have $\bar{\mathcal{J}}_{\mathcal{S}} \leq \bar{\mathcal{J}}_{\mathcal{B}}$. The actual noise ratio satisfies:

$$\left.\frac{\mathbb{E}[\|g^{\mathcal{S}}_{\text{noise}}\|^2]}{\mathbb{E}[\|g^{\mathcal{B}}_{\text{noise}}\|^2]}\right|_{\text{actual}} \leq \frac{|\mathcal{B}|}{|\mathcal{S}|}\cdot\frac{\bar{r}_{\mathcal{S}}}{\bar{r}_{\mathcal{B}}}\cdot\underbrace{\frac{\bar{\mathcal{J}}_{\mathcal{S}}}{\bar{\mathcal{J}}_{\mathcal{B}}}}_{\leq 1} \leq \frac{|\mathcal{B}|}{|\mathcal{S}|}\cdot\frac{\bar{r}_{\mathcal{S}}}{\bar{r}_{\mathcal{B}}} \tag{27}$$

The additional factor $\bar{\mathcal{J}}_{\mathcal{S}}/\bar{\mathcal{J}}_{\mathcal{B}} \leq 1$ quantifies the *dual noise reduction*—filtering high-$\|\varepsilon_i\|$ samples simultaneously removes high-$\|J_i\|_F$ samples.

D.3 Capacity Reallocation

Proposition 1 also explains the difficulty-dependent behavior observed in the tail error analysis (Section 4.3): by suppressing noisy gradients from unreliable teacher predictions, selective distillation reallocates the student's learning capacity from over-fitting potentially misleading signals to focusing on samples where the teacher provides genuinely informative guidance. The result is a principled trade-off: slight degradation on easy samples (where the default-threshold configuration already performs well) in exchange for substantial improvement on hard samples (where stricter filtering provides the most valuable corrective signal).

D.4 Threshold Form Justification

The threshold form $\tau = \mu_c + \kappa \cdot \sigma_c$ is *distribution-agnostic* in the sense that it defines a relative filtering strictness via empirical statistics regardless of the score distribution shape. When $\{c_i\}$ are approximately normally distributed, the retention rate admits a closed-form expression: denoting the standard normal CDF by Φ, the fraction of retained samples is $1 - \Phi(\kappa) = \Phi(-\kappa)$ (by the symmetry of the Gaussian). However, normality is not required for the mechanism to be effective—κ controls filtering in units of standard deviations for any distribution with finite variance, and Chebyshev's inequality guarantees that at most $1/\kappa^2$ of the population lies beyond κ standard deviations from the mean. The hybrid threshold $\tau = \beta \cdot \tau_{local} + (1-\beta) \cdot \tau_{global}$ further addresses distributional non-stationarity: the global component provides stable, dataset-level calibration, while the local component adapts to batch-level distributional shifts during training.

APPENDIX E
RELIABILITY-AWARE AND SELECTIVE LEARNING

The reliability-aware selection mechanism in EDG++ relates to several lines of research on selective and robust learning. Curriculum learning [52] and self-paced learning [53] train models by presenting samples in a meaningful order, typically from easy to hard. MentorNet [71] learns a data-driven curriculum for training on corrupted labels. Co-teaching [54] trains two networks simultaneously, with each network selecting small-loss samples for the other. More broadly, learning from noisy labels [55] encompasses a family of methods that identify and handle unreliable supervision signals.

Our approach differs from these methods in a fundamental way: we estimate *teacher reliability* rather than sample difficulty or label noise. In standard noisy label settings, the ground-truth labels are corrupted and the goal is to identify clean samples. In our setting, the ground-truth task labels are clean, but the auxiliary distillation signal from the teacher varies in quality across samples. This distinction motivates our design choice of a separately pre-trained reliability estimator that predicts the teacher's reconstruction error, combined with an adaptive thresholding mechanism that balances global consistency with local adaptivity. The confidence calibration literature [47] also informs our approach, as well-calibrated confidence scores are essential for effective threshold-based filtering.

APPENDIX F
DIFFICULTY-STRATIFIED ERROR ANALYSIS

Figure S1 stratifies test samples into six difficulty percentiles and plots the relative per-bin error change for all four representative tasks. The picture is striking: every task traces a nearly identical monotonic descent from degradation on easy samples to improvement on hard samples, crossing zero near P50–75. In relative terms, easy samples (P0–25) suffer

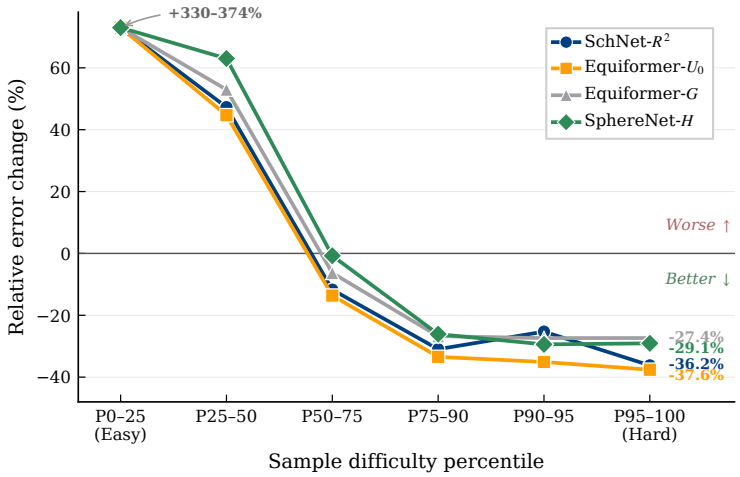

Fig. S1: Difficulty-stratified relative error change on QM9. All four tasks are plotted on a single panel; the y-axis shows the per-bin relative change in mean absolute error between the tuned-κ and default-κ configurations. Hard samples (P75–100) show consistent improvement of 25–38%, while the crossover near P50–75 and the remarkably similar trajectories across architectures confirm that capacity reallocation is a general mechanism of adaptive thresholding. *Note*: P0–25 values are truncated at the y-axis limit; actual relative degradation on easy samples ranges from +330% to +374%, but corresponds to negligible absolute error increases.

a large increase (+330–374%), whereas hard samples (P95–100) achieve a 27–38% reduction. Because the absolute errors of easy samples are near-zero under the default-threshold configuration, the large relative increase translates to a negligible absolute penalty; conversely, the moderate relative decrease on hard samples corresponds to a substantial absolute gain.

This difficulty-dependent behavior is the intended mechanism of threshold tuning: by raising κ beyond the default, EDG++ applies stricter filtering that suppresses potentially misleading ED signals on samples where the default-threshold model already performs well, reallocating model capacity to under-optimized hard samples. The remarkable consistency of this crossover pattern across three architectures and four quantum properties provides strong evidence that capacity reallocation is a general mechanism of adaptive thresholding.

Robustness metrics further confirm variance reduction: standard deviation reductions of 5.3% (SchNet-R^2), 16.6% (Equiformer-U_0), 17.3% (Equiformer-G), and 11.1% (SphereNet-H). Maximum error reductions are pronounced for Equiformer: 39.1% (U_0) and 37.9% (G). For Equiformer-U_0, the maximum error decreases from 1.665 to 1.014, a 39% reduction that substantially enhances reliability in practical applications.

Appendix G
Property-Type Grouped Analysis

Table S1 provides a detailed property-type grouped analysis of EDG++ improvements.

TABLE S1: Property-type grouped analysis of EDG++ improvement over the no-distillation baseline on QM9. "Avg Δ" denotes the mean relative improvement. "Win" denotes the number of properties improved.

Group	SchNet		Equiformer		SphereNet	
	Avg Δ	Win	Avg Δ	Win	Avg Δ	Win
Electronic	+2.24%	5/5	+3.64%	4/5	+1.96%	5/5
Thermo.	+3.70%	5/5	+16.30%	5/5	+5.52%	5/5
Vibrational	+2.30%	1/1	−3.20%	0/1	+4.40%	1/1
Structural	+6.40%	1/1	+11.90%	1/1	+1.90%	1/1
Overall	+3.20%	12/12	+9.03%	10/12	+3.64%	12/12

Appendix H
Distillation Weight Sensitivity

Table S2 reports the per-task λ sensitivity under the default-threshold configuration.

TABLE S2: λ sensitivity (relative performance difference between best and worst $\lambda \in \{0.01, 0.5, 1.0\}$) under the default-threshold configuration ($\kappa_{batch} = \kappa_{all} = 0$, $\beta = 0$).

Task	SchNet	Equiformer	SphereNet
α	1.19%	1.21%	3.52%
C_v	1.09%	1.34%	1.89%
G	2.37%	12.85%	4.59%
$\Delta\varepsilon$	0.71%	1.56%	1.74%
H	0.33%	5.62%	6.88%
$\varepsilon_{\text{HOMO}}$	1.79%	1.75%	0.92%
$\varepsilon_{\text{LUMO}}$	1.39%	1.62%	2.35%
μ	1.20%	3.10%	1.94%
R^2	2.21%	15.26%	5.84%
U_0	1.94%	34.87%	7.34%
U	1.26%	11.06%	4.35%
ZPVE	2.21%	3.00%	0.96%
Average	**1.47%**	**7.77%**	**3.53%**

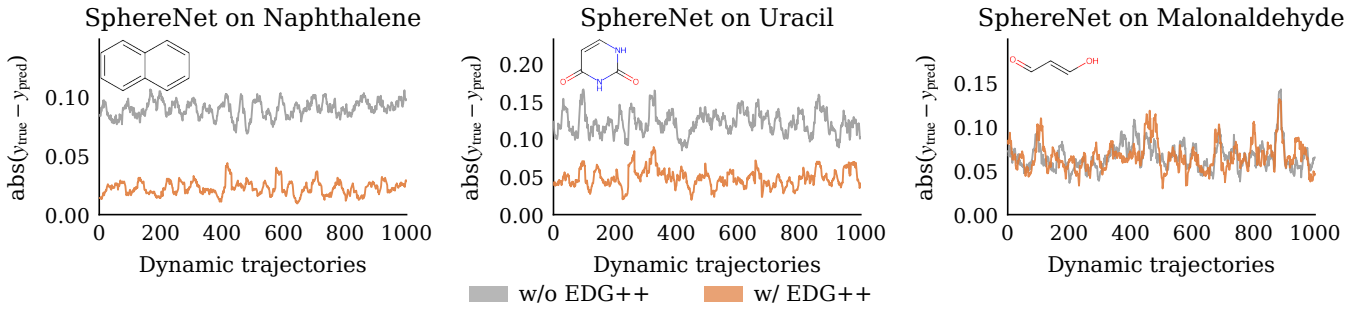

Fig. S2: Per-sample energy prediction errors on rMD17 (SphereNet), comparing the no-distillation baseline with EDG++ ($\kappa_{all} = 1.5$ for naphthalene and uracil; $\kappa_{all} = 1.0$ for malonaldehyde). EDG++ consistently reduces errors on naphthalene ($\Delta = +74.9\%$) and uracil ($\Delta = +60.4\%$); malonaldehyde shows nearly overlapping curves ($\Delta = -0.2\%$). *Note*: Δ values are computed from per-sample predictions of independently trained baseline and EDG++ models, not from the conference-version baselines in Tables 3–4.

APPENDIX I
rMD17 PREDICTION ERROR VISUALIZATION

APPENDIX J
HYPERPARAMETER INTERACTION ANALYSIS

Figure S3 visualizes the relative improvement over the baseline as a function of κ_{batch} and κ_{all} under the hybrid strategy ($\beta = 0.5$) for four representative tasks. Two observations emerge. First, the improvement landscape is generally smooth, with most cells showing positive gains, indicating that EDG++ is robust to the specific choice of κ values. Second, the optimal region varies by task and model: for SchNet, the improvements are uniformly distributed (all cells within 1% of each other), while for SphereNet, thermodynamic properties like G show more pronounced sensitivity to κ_{all}, with stricter filtering ($\kappa_{all} = 1.5$) yielding larger gains.

Threshold sensitivity on rMD17. We further analyze how κ_{all} affects energy prediction on rMD17 (SphereNet, $\lambda = 0.001$). The sensitivity patterns are highly molecule-dependent: aspirin and toluene favor negative κ_{all} (more inclusive filtering), while naphthalene and uracil favor positive κ_{all} (stricter filtering). Malonaldehyde is the most extreme case, with energy improvement fluctuating between -260% and $+10\%$ across $\kappa_{all} \in [-1.5, 1.5]$, consistent with its challenging proton transfer dynamics. Force predictions are relatively insensitive to κ_{all} across all molecules, suggesting that the ED teacher primarily provides global energy-level information rather than local force-level gradients.

APPENDIX K
IMAGEED RECONSTRUCTION VISUALIZATION

Figure S4 shows the reconstruction quality of ImageED. ImageED can faithfully reconstruct ED images compared to the originals, indicating that it effectively learns ED-related knowledge from the multi-view RGB-D images. Removing the masked reconstruction task (ImageED w/o $\mathcal{L}_{MR}$) degrades the reconstruction of local pixel-level details, confirming the importance of both pretext tasks.

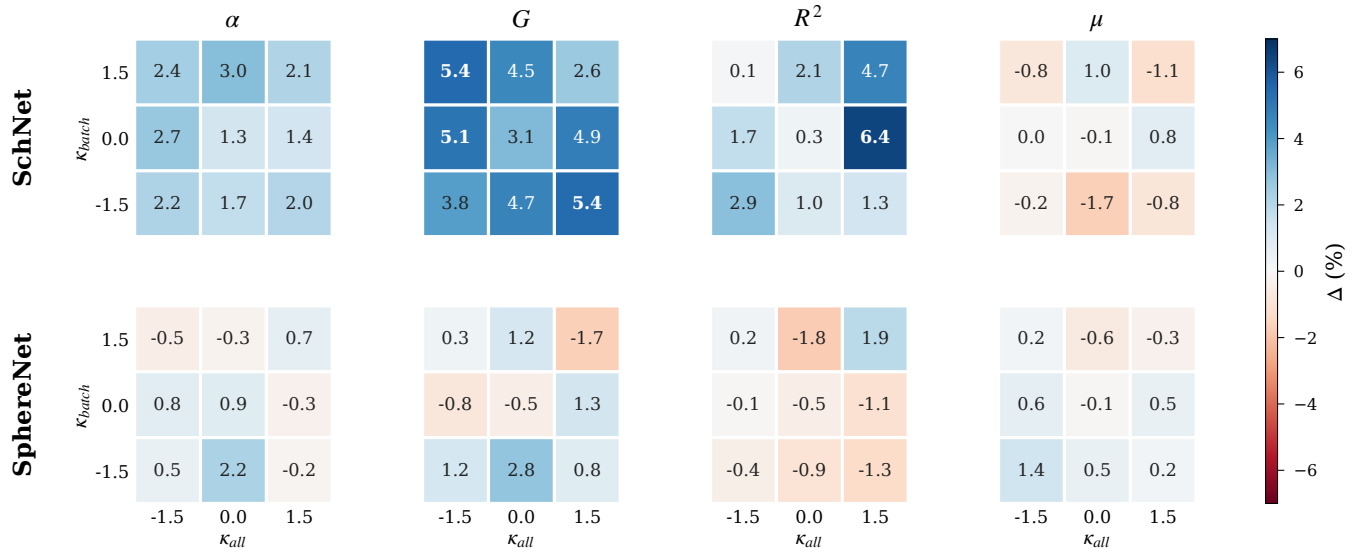

Fig. S3: Relative improvement (%) over the no-distillation baseline as a function of κ_{batch} and κ_{all} with $\beta = 0.5$ on QM9. For each (κ_{batch}, κ_{all}) pair, we report the best result across all λ values. Blue indicates improvement; red indicates degradation; white corresponds to no change. Bold values denote $|\Delta| > 5\%$. *Note*: color scales are independently normalized per panel to maximize contrast within each task-model combination.

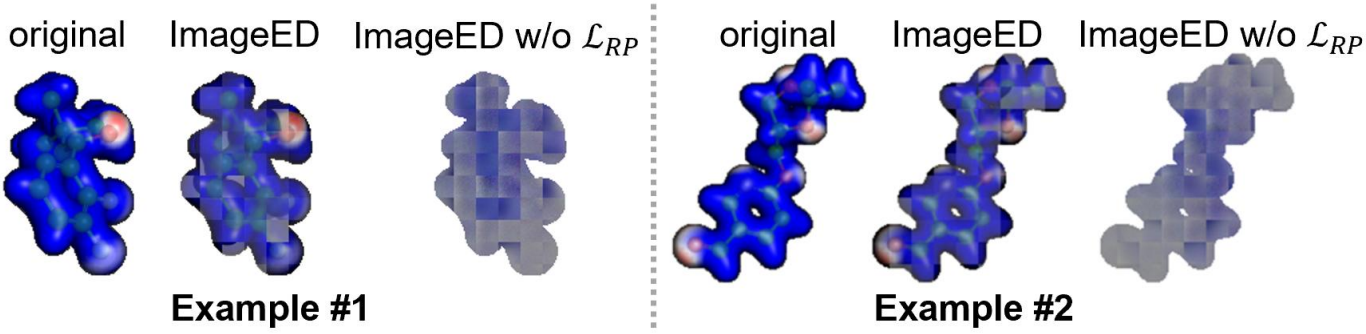

Fig. S4: Visualization of ImageED reconstruction. ImageED generates high-quality ED images compared to the originals. Removing the masked reconstruction task ($\mathcal{L}_{MR}$) limits the understanding of local pixel-level features.

Appendix L
ED Image Representation Comparison

TABLE S3: RMSE performance of different ED representations on 6 energy prediction tasks. Point cloud, voxel, and image use PointNet, ResNet3D, and ResNet18, respectively. E1–E6 represent DF-RKS Final Energy, Nuclear Repulsion Energy, One-Electron Energy, Two-Electron Energy, DFT Exchange-Correlation Energy, and Total Energy.

Repr.	E1	E2	E3	E4	E5	E6
Point	275.1	168.6	557.7	244.8	14.5	288.9
Voxel	121.1	313.9	947.8	271.2	7.9	202.0
Image	**111.6**	**47.9**	**349.8**	**85.6**	**4.4**	**124.5**
Δ	↑7.8%	↑71.6%	↑37.3%	↑65.0%	↑44.5%	↑38.4%

Appendix M
Computational Cost Analysis

EDG++ introduces no additional parameters or forward passes beyond EDG; the only overhead is a scalar reliability score lookup per sample, resulting in <5% per-epoch overhead compared to the no-distillation baseline. All experiments were conducted on 8× NVIDIA H200 GPUs. On QM9, a single distillation run takes ∼18h (SchNet, 1000 epochs), ∼35h

(SphereNet, 1000 epochs), or ∼40h (Equiformer, 300 epochs). The dominant cost is the grid search: each model requires 45 configurations (3 λ values $\times$ 15 (κ, β) combinations) per task, totaling ∼9.6k, ∼19.0k, and ∼21.3k GPU-hours for SchNet, SphereNet, and Equiformer, respectively. On rMD17, SphereNet requires 7 κ_{all} values per molecule (∼46h per run), totaling ∼3.1k GPU-hours. In practice, the search can be substantially reduced by first identifying a good λ on a representative task and then only searching over (κ, β), which would reduce the search space by 3×.

APPENDIX N
CONTRIBUTION DECOMPOSITION AND BASELINE DISCUSSION

Our experimental design provides a natural two-level decomposition: Baseline → EDG isolates the contribution of ED as privileged information, and EDG → EDG++ isolates the contribution of reliability-aware selective distillation.

Why standard KD baselines are not included. We acknowledge that comparisons with adapted standard KD baselines (*e.g.*, FitNets [42], AT [43]) would strengthen the paper. In principle, one could implement a "structural-image teacher → geometry student" feature matching control without ED pre-training, which would help disentangle whether the gains come from (1) the ED signal itself, (2) the auxiliary teacher architecture, or (3) the cross-modal training paradigm. We did not include such controls due to computational budget constraints: the full EDG++ grid search already required ∼50k GPU-hours.

However, several lines of indirect evidence support that the gains primarily originate from the ED signal rather than the teacher architecture *per se*: (i) the improvement hierarchy across property types (thermodynamic > electronic > vibrational) is physically consistent with the Hohenberg-Kohn theorem—a generic teacher architecture would not produce this pattern; (ii) the improvements are consistent across four fundamentally different student architectures (SchNet, Equiformer, SphereNet, ViSNet), suggesting the transferred knowledge is architecture-agnostic; (iii) the reliability estimator, trained to predict *ED reconstruction error*, is effective at identifying samples where distillation helps—this would not work if the gains were unrelated to ED quality. We encourage future work to include a non-ED teacher control to formally validate this decomposition.

APPENDIX O
DATASET DESCRIPTIONS

O.1 QM9 Dataset

QM9 [66] is a widely used benchmark for quantum chemistry, containing 133,885 small organic molecules with up to 9 heavy atoms (C, N, O, F). Each molecule is associated with 12 quantum mechanical properties computed using density functional theory (DFT) at the B3LYP/6-31G(2df,p) level. Table S4 provides detailed descriptions of these properties.

TABLE S4: Descriptions of 12 quantum mechanical properties in QM9.

Property	Description	Unit	Type
α	Isotropic polarizability	a_0^3	Electronic
$\Delta\varepsilon$	HOMO-LUMO gap	meV	Electronic
ε_{HOMO}	Highest occupied molecular orbital energy	meV	Electronic
ε_{LUMO}	Lowest unoccupied molecular orbital energy	meV	Electronic
μ	Dipole moment	D	Electronic
C_v	Heat capacity at 298.15K	$\frac{cal}{mol \cdot K}$	Thermodynamic
G	Free energy at 298.15K	meV	Thermodynamic
H	Enthalpy at 298.15K	meV	Thermodynamic
R^2	Electronic spatial extent	a_0^2	Structural
U	Internal energy at 298.15K	meV	Thermodynamic
U_0	Internal energy at 0K	meV	Thermodynamic
ZPVE	Zero point vibrational energy	meV	Vibrational

O.2 rMD17 Dataset

The revised MD17 (rMD17) [67] dataset contains molecular dynamics trajectories for 10 small organic molecules. Each trajectory consists of molecular conformations with corresponding energies and atomic forces computed at the PBE+vdW-TS level of theory. Table S5 provides detailed descriptions of these molecules.

APPENDIX P
PAIRED BOOTSTRAP CONFIDENCE INTERVALS

To quantify the estimation uncertainty of our single-seed results, we compute paired bootstrap 95% confidence intervals based on per-sample test predictions (10,831 samples for QM9). For each task-model pair, we resample the test set 10,000 times with replacement and compute the MAE on each resample. Table S6 reports the bootstrap CI for representative tasks.

The mean relative CI width across all 36 task-model pairs is 7.0% (median 6.1%, max 12.5%), indicating that the single-seed point estimates are reasonably stable. The paired bootstrap tests confirm that several key improvements (Equiformer-U_0, SphereNet-ZPVE) are statistically robust even within a single seed ($p < 0.01$), while others (SphereNet-H) require multi-seed evaluation for definitive conclusions.

TABLE S5: Descriptions of 10 molecules in rMD17.

No.	Molecule	Formula	#Atoms	#Conformations
1	Aspirin	$C_9H_8O_4$	21	211,762
2	Azobenzene	$C_{12}H_{10}N_2$	24	99,999
3	Benzene	C_6H_6	12	627,983
4	Ethanol	C_2H_6O	9	555,092
5	Malonaldehyde	$C_3H_4O_2$	9	993,237
6	Naphthalene	$C_{10}H_8$	18	326,250
7	Paracetamol	$C_8H_9NO_2$	20	106,490
8	Salicylic acid	$C_7H_6O_3$	16	320,231
9	Toluene	C_7H_8	15	442,790
10	Uracil	$C_4H_4N_2O_2$	12	133,770

TABLE S6: Bootstrap 95% CIs for EDG++ (per-task-best) and paired bootstrap test of EDG++ vs. EDG on representative QM9 tasks. Δ: paired difference in MAE (EDG $-$ EDG++; positive means EDG++ is better).

Model	Task	EDG++ MAE [95% CI]	Δ [95% CI]	p
SchNet	R^2	0.126 [0.121, 0.132]	+0.003 [+0.000, +0.006]	0.020
Equif.	U_0	0.017 [0.016, 0.018]	+0.001 [+0.001, +0.002]	<0.001
Equif.	μ	0.021 [0.020, 0.022]	0.000	n.s.
Sphere.	H	0.006 [0.006, 0.007]	+0.000 [−0.000, +0.000]	0.472
Sphere.	ZPVE	0.001 [0.001, 0.001]	+0.000 [+0.000, +0.000]	0.003

APPENDIX Q
HYPERPARAMETER SEARCH SPACE

We conduct grid search over the following hyperparameters for EDG++:

- **Distillation weight** $\lambda \in \{0.01, 0.5, 1.0\}$ for QM9; $\lambda = 0.001$ for rMD17
- **Local threshold coefficient** $\kappa_{batch} \in \{-1.5, 0, 1.5\}$
- **Global threshold coefficient** $\kappa_{all} \in \{-1.5, 0, 1.5\}$ for QM9; $\{-1.5, -1, -0.5, 0, 0.5, 1.0, 1.5\}$ for rMD17
- **Mixing coefficient** $\beta \in \{0, 0.5, 1.0\}$ for QM9; $\beta = 0$ for rMD17

For each task, we select the hyperparameter configuration that achieves the best validation performance.

APPENDIX R
IMPLEMENTATION DETAILS

R.1 ImageED Pre-training

The encoder and decoder of ImageED are built based on ViT-Large/16 [62]. The encoder has an embedding dimension of 1024 with 24 Transformer layers, and the decoder has an embedding dimension of 512 with 8 Transformer layers. We use a learning rate of 1.5e-4, a batch size of 64, a mask ratio of 0.75, $\lambda_{VR} = 1$ and $\lambda_{MR} = 1$ for 20 epochs on 8 GeForce RTX 4090 GPUs. The input ED images have shape $6 \times 4 \times 224 \times 224$ (6 views, 4 channels including RGB-D).

R.2 ED-aware Teacher Pre-training

We divide 2% of the 2 million molecules as validation set and the rest as training set. The ED-aware teacher uses ResNet18 [63] with view-wise average pooling. The ED predictor is a 2-layer MLP with Softplus activation. We use a learning rate of 5e-3 and a batch size of 128 to train for approximately 280k steps.

R.3 Reliability Estimator Training

The reliability estimator is a lightweight MLP with architecture: Linear(512, 256) $\rightarrow$ Softplus $\rightarrow$ Linear(256, 1), with approximately 131K parameters. It is jointly trained with the ED-aware teacher and ED predictor in Stage II, sharing the same SGD optimizer (momentum 0.9, learning rate 0.003, weight decay 1e-4, batch size 128, 25 epochs). Note that the evaluator's input features are detached from the teacher's computation graph to prevent gradient interference. The training objective is to predict the L2 reconstruction error between predicted and ground-truth ED features.

R.4 Downstream Task Training

For QM9, we use the dataset split from Geom3D [69]: 110K for training, 10K for validation, and 11K for testing. For rMD17, we use 950 for training, 50 for validation, and 1000 for testing. The mapper f_M and task predictor f_T each consist of a single linear layer. We select the best checkpoint based on validation MAE for QM9 and validation force MAE for rMD17.